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Chapter 1: State feedback design using pole placement and LQR

The system considered in this homework is the cart-pendulum balance system. The objective is to stabilize the pendulum around the upright equilibrium while controlling the horizontal motion of the cart. The model contains four main state variables: cart position p, pendulum angle theta, cart velocity p_dot and angular velocity theta_dot.

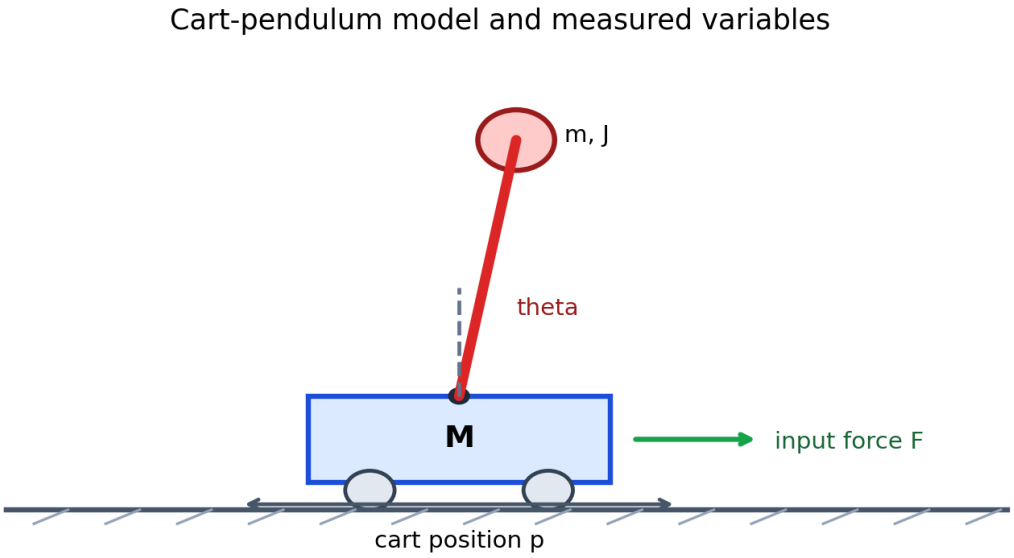


Figure 1. Cart-pendulum model used for the balance-control problem.

A compact representation of the nonlinear dynamics is written as

```
M(q) q_ddot + C(q, q_dot) + K(q) = B(q) u
```

where M(q) is the inertia matrix, C(q, q_dot) contains damping and velocity-dependent terms, K(q) represents the gravity-related term, and u = F is the force applied to the cart. Linearization is carried out around theta = 0, with sin(theta) approximately theta and cos(theta) approximately 1.

With x = [p, theta, p_dot, theta_dot]^T, the linear state-space model is

```
x_dot = A x + B u,      y = C x
```

```
mu = Mt Jt - m^2 l^2 = 9800
```

For the numerical example, the following parameters are used.

Parameter	Value
M	10 kg
m	80 kg
J	100 kg m^2
l	1 m
c	0.1 N s/m
gamma	0.01 N m s
g	9.8 m/s^2

The matrices obtained from these parameters are

```
[[ 0.      0.      1.      0.    ]
 [ 0.      0.      0.      1.    ]
 [ 0.      6.4    -0.0018 -0.0001]
 [ 0.      7.2    -0.0008 -0.0001]]
```

```
B = [0.      0.      0.01837 0.00816]
```

The open-loop system has one unstable pole, which confirms that a feedback controller is required to stabilize the upright position. Since the reachability matrix has full rank, arbitrary closed-loop pole assignment is possible.

```
R = [B  AB  A^2B  A^3B],     rank(R) = 4
```

1.1 Ackermann pole-placement controller

The desired closed-loop poles are selected as two complex-conjugate pairs in the left half-plane:

```
lambda = {-0.33 +/- 0.66j,  -0.18 +/- 0.18j}
```

The corresponding fourth-order polynomial is

```
p*(s) = s^4 + 1.02 s^3 + 0.8469 s^2 + 0.238788 s + 0.0352836
```

Using Ackermann formula, the state-feedback vector, with the sign convention u = Kx, is

```
K_Ack = [0.4410,  -986.7379,  3.0848,  -131.6546]
```

This places the eigenvalues of A + BK at the desired locations. In a Simulink negative-feedback implementation, the equivalent gain sign must be adjusted consistently.

1.2 LQR controller

The LQR method introduces a quadratic cost that penalizes both state deviations and control effort:

```
J = integral_0^infinity (x^T Q x + u^T R u) dt
```

Bryson-type scaling is used to choose the diagonal weights according to acceptable values of p, theta, p_dot and theta_dot. Solving the Riccati equation gives the following stabilizing gain in the convention u = Kx:

```
K_LQR = [7.9056,  -2070.3716,  44.0014,  -807.9465]
```

The closed-loop poles are approximately

```
lambda_1,2 = -2.6834 +/- 0.0262j,     lambda_3,4 = -0.2111 +/- 0.2079j
```

The LQR poles contain a fast pair and a slower pair. The slow pair dominates the final settling, while the fast pair quickly damps the more aggressive pendulum motion.

Chapter 2: Comparison of the two controllers

To compare the pole-placement controller and the LQR controller, a unit step reference is applied to the cart position. A feedforward coefficient kr is included so that the steady-state cart position matches the reference.

```
u = Kx + kr r,      kr = -1 / [C(A + BK)^(-1)B],      C = [1 0 0 0]
```

For the selected gains, kr is -0.4410 for the Ackermann design and -7.9056 for the LQR design.

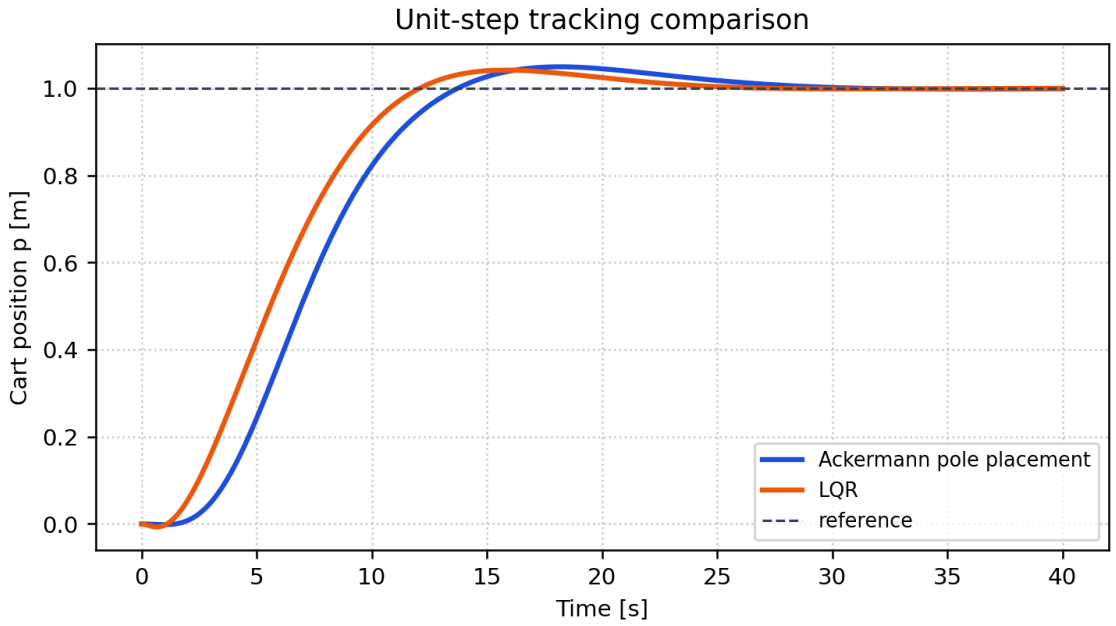


Figure 2. Unit-step response of the cart position for the two feedback designs.

Both controllers stabilize the system and track the reference. The LQR response is slightly faster in the main transient and has a small improvement in overshoot, while the Ackermann design follows the pole-placement specification more directly.

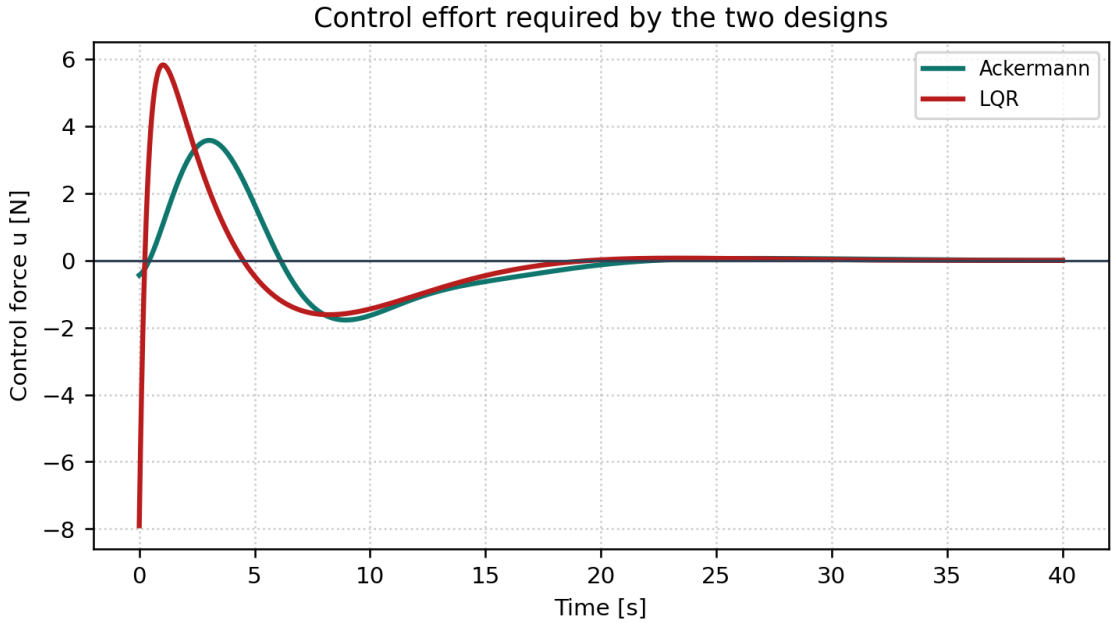


Figure 3. Control effort comparison for the pole-placement and LQR designs.

The LQR controller requires a larger initial control action. This is expected because its gain is chosen through an optimization problem that strongly penalizes state deviations, especially pendulum angle errors. The Ackermann controller gives a more moderate force peak for this particular pole selection.

2.1 Region of attraction discussion

For the nonlinear system, a local region of attraction can be estimated through a Lyapunov-level-set argument. A quadratic candidate V(x) = x^T P x is evaluated along the nonlinear closed-loop dynamics. The largest tested level set for which V_dot remains negative gives an inner approximation of the true region of attraction.

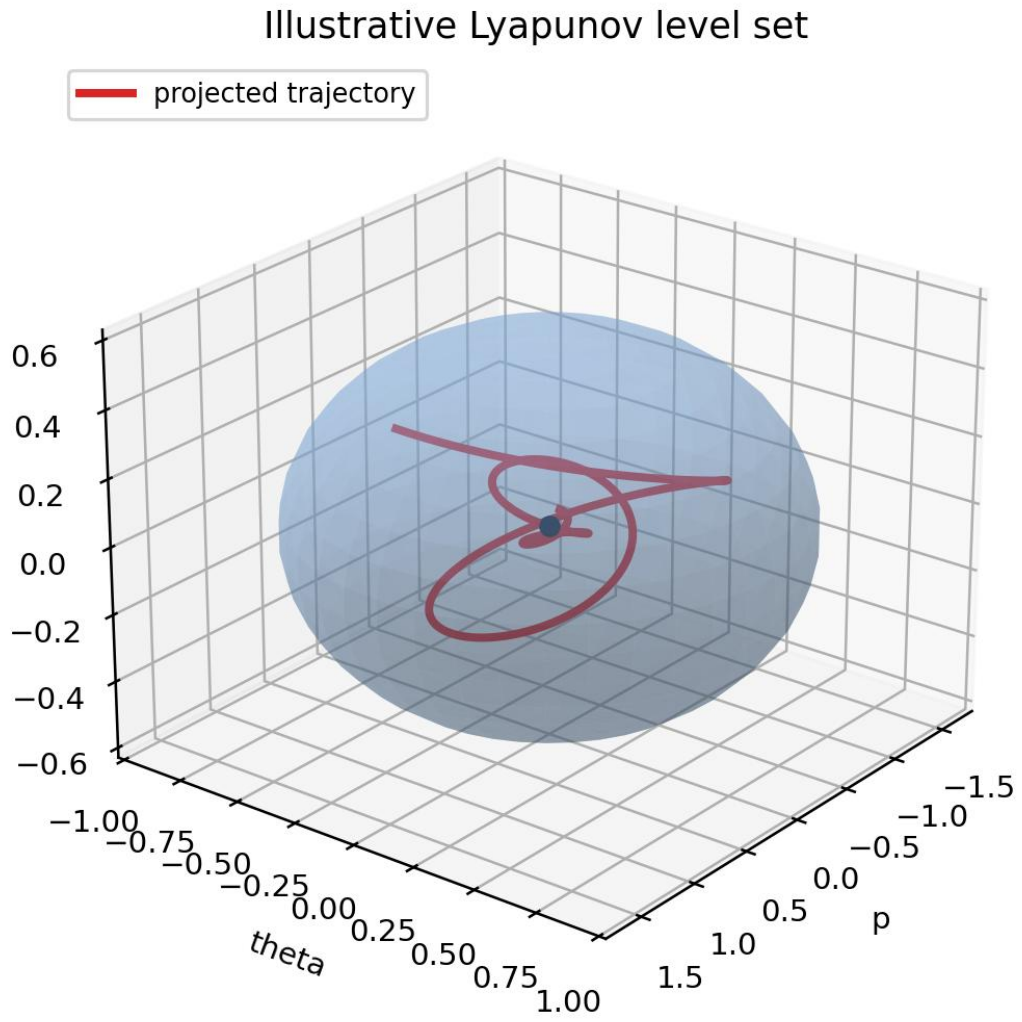


Figure 4. Conceptual Lyapunov ellipsoid and projected trajectory toward the equilibrium.

Using the numerical procedure, the certified scalar levels were approximately c = 0.057 for the Ackermann design and c = 1.388 for the LQR design. Since these values correspond to ellipsoidal level sets, they should be interpreted as relative indicators rather than exact geometric volumes. Nevertheless, the larger value for the LQR design indicates a wider certified region and better tolerance to nonlinear perturbations.

Chapter 3: Transfer functions, stability and minimum-phase properties

The same linearized model can be used to compute transfer functions from the cart force F to the two measured outputs, namely cart position p and pendulum angle theta.

For the transfer function from force to cart displacement, the output matrix is C_p = [1 0 0 0]. The resulting transfer function is

```
G_Fp(s) = (18000 s^2 + 10 s - 784000) / (9800000 s^4 + 18900 s^3 - 70559999 s^2 - 78400 s)
G_Fp(s) approximately = 0.01836 (s + 2.0870)(s - 2.0869) / [s(s + 0.0011)(s + 2.6836)(s - 2.6828)]
```

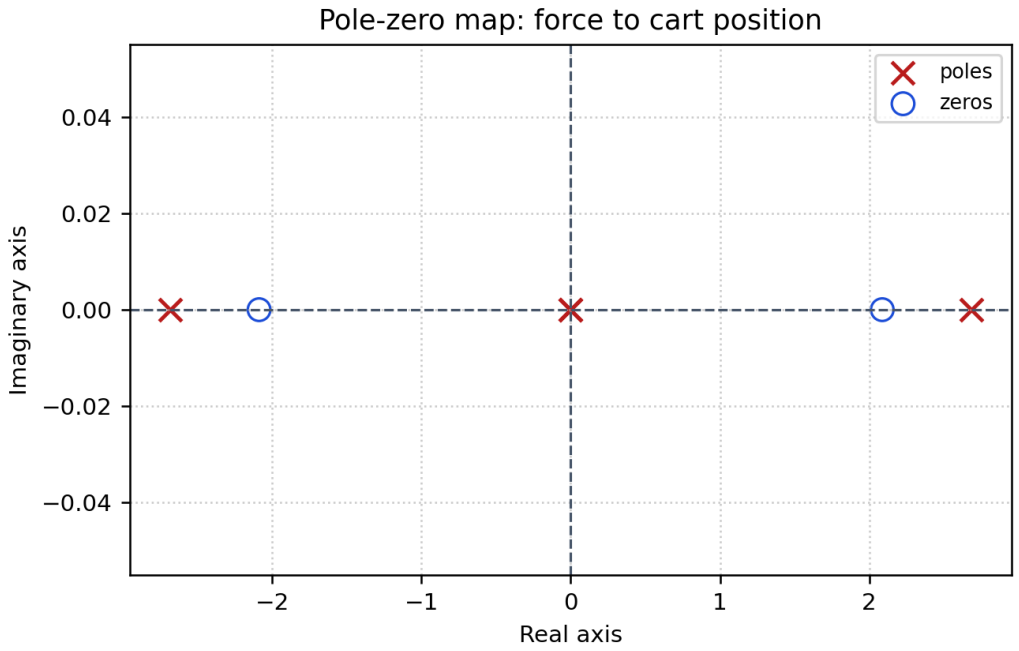


Figure 5. Pole-zero map of the force-to-position transfer function.

This transfer function contains an unstable pole and a right-half-plane zero. Therefore, it is unstable and non-minimum phase. The relative degree is 2, because the denominator order is four and the numerator order is two.

For the transfer function from force to pendulum angle, the output matrix is C_theta = [0 1 0 0]. After simplifying the pole-zero cancellation at the origin,

```
G_Ftheta(s) = 2s / [245 (s + 2.6836)(s - 2.6828)(s + 0.0011)]
```

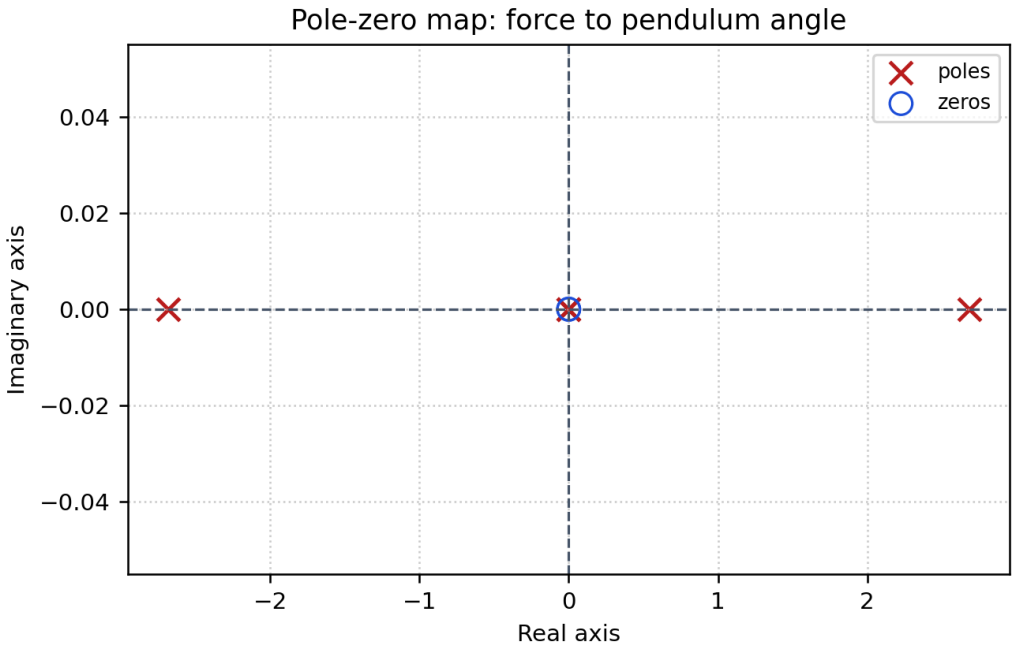


Figure 6. Pole-zero map of the force-to-angle transfer function.

The angle transfer function also contains the unstable pole at approximately +2.6828. The cancellation at s = 0 is related to loss of observability when only the angular displacement is used as the output. For this reason, designing a controller only through this SISO transfer function is not as reliable as using the full state-space model.

Transfer function	Stability	Minimum phase	Relative degree	Important feature
F -> p	Unstable	No	2	RHP pole and RHP zero
F -> theta	Unstable	No	2	RHP pole and zero-pole cancellation at the origin

Chapter 4: Position tracking while maintaining balance

The final task is to move the cart from one position to another while keeping the pendulum close to the upright equilibrium. In addition to state feedback, an integral state is introduced to remove steady-state position error.

```
eta_dot = p - p_ref
```

The augmented state is

```
x_a = [p - p_ref, theta, p_dot, theta_dot, eta]^T
```

The augmented LQR gain used in the simulation is

```
K_a = [-32.6979,  2403.6449,  -95.5994,  970.4546,  -5.2704]
```

The implemented control law is written explicitly as

```
u = -K_a x_a
```

A saturation block of +/-80 N is included to keep the force in a realistic actuator range. The integrator term prevents permanent position error, while the state-feedback terms stabilize the pendulum angle.

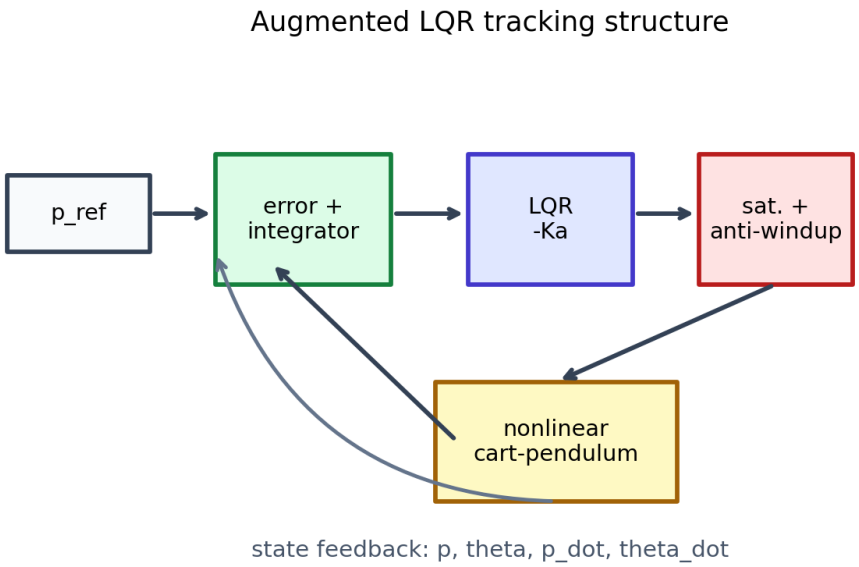


Figure 7. Augmented LQR structure with position-error integration and actuator saturation.

The nonlinear cart-pendulum equations are simulated with the initial condition p(0) = 5 m, theta(0) = 0, p_dot(0) = 0 and theta_dot(0) = 0. At t = 1 s, the reference changes from 5 m to 8 m.

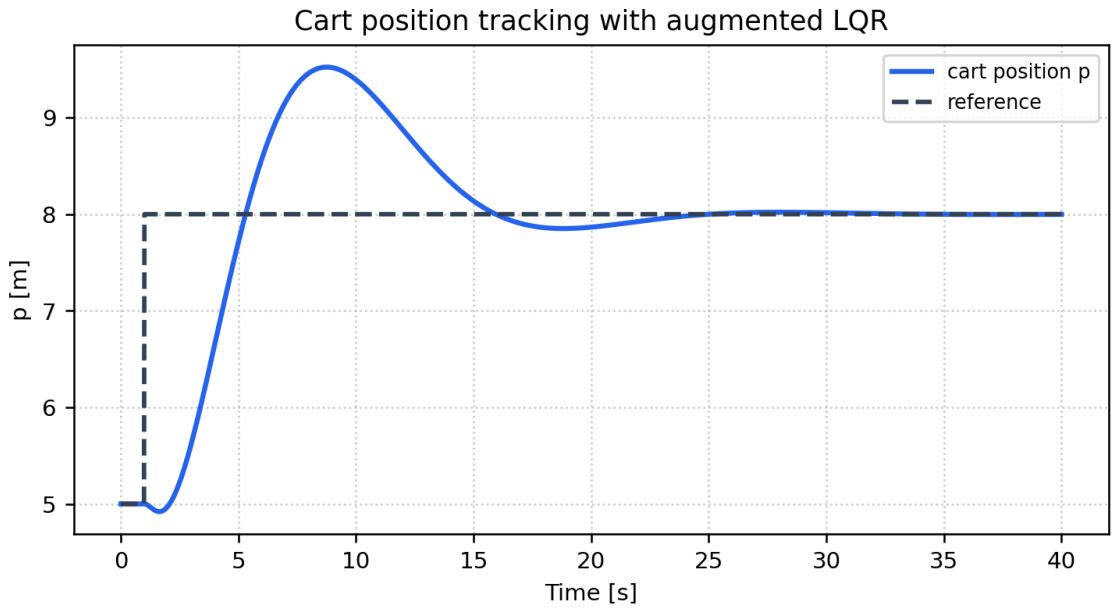


Figure 8. Cart position response for the nonlinear model with augmented LQR.

The cart reaches the new reference without a steady-state error. The response is intentionally not extremely fast, because excessive acceleration would produce a large pendulum deflection and large actuator demand.

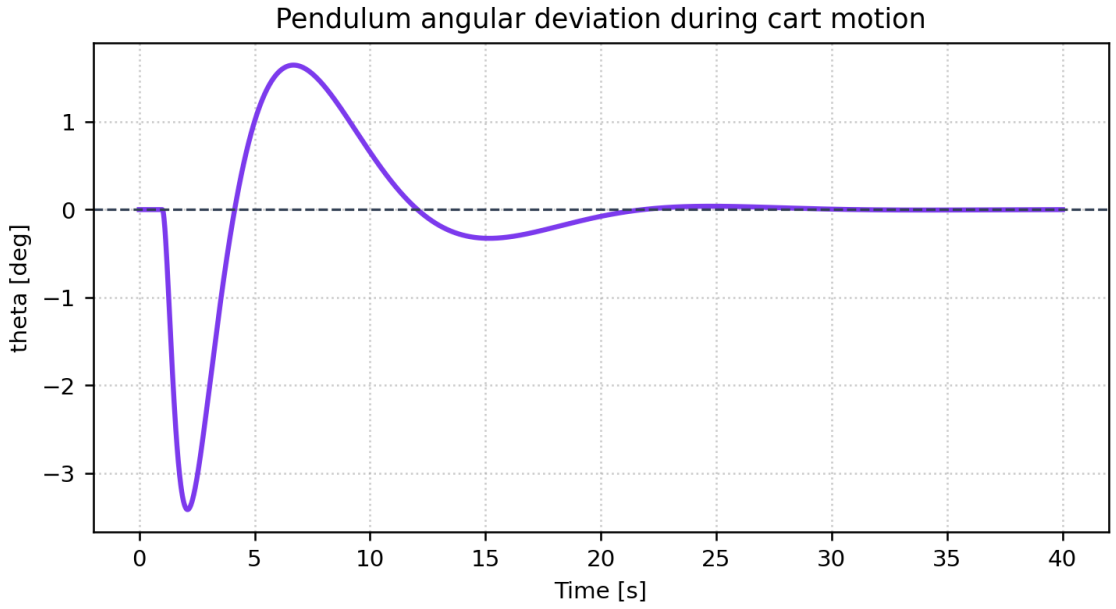


Figure 9. Pendulum angular displacement during the position transition.

The pendulum angle deviates only temporarily and then returns to the upright equilibrium. This confirms that the controller is not only translating the cart, but also preserving balance.

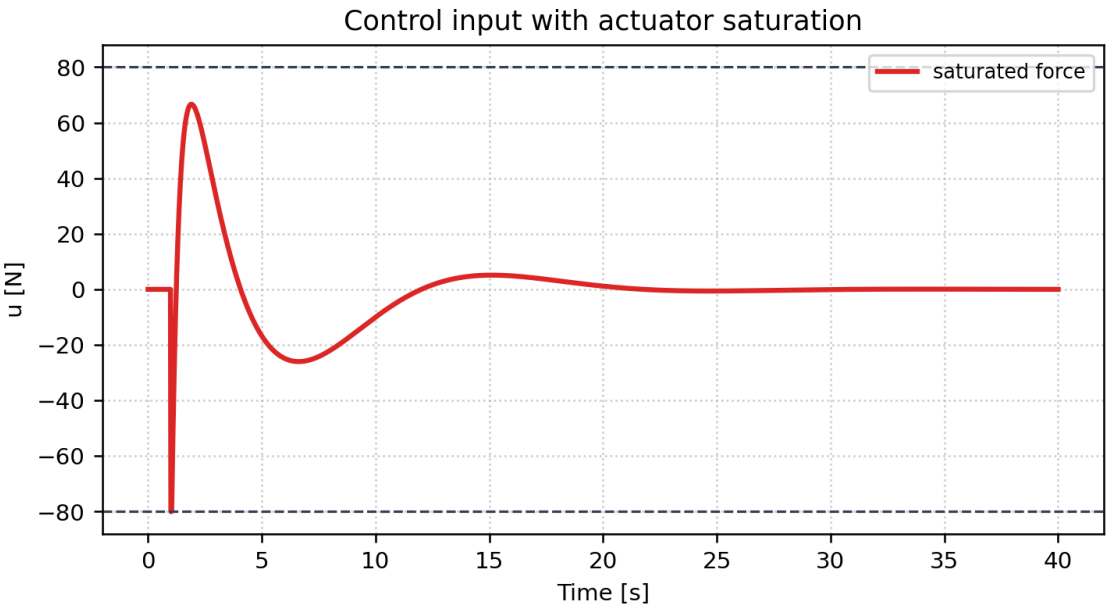


Figure 10. Saturated control force used during the tracking manoeuvre.

The control signal stays within the selected saturation limits. The first force peak is caused by the step change in desired cart position; after the transient, the required force converges close to zero.

Conclusion

The cart-pendulum system is unstable around the upright equilibrium and requires active feedback stabilization. Pole placement gives direct control of the closed-loop eigenvalues, while LQR provides a systematic compromise between transient quality and control effort. The transfer-function analysis shows that the open-loop plant is unstable and non-minimum phase, which explains why full state feedback is a more suitable framework. Finally, the augmented LQR design with integral action succeeds in moving the cart between two positions while keeping the pendulum balanced and maintaining the control force within realistic bounds.